PAPER_758: Rings of Relativity — Einstein Ring GAL-CLUS-022058s UQFF Lensing

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Abstract

GAL-CLUS-022058s (the "Molten Ring") is one of the largest Einstein rings discovered by Hubble, with a lens cluster at z ≈ 0.5 and source at z ≈ 1.0. This paper derives the UQFF-modified Einstein-ring gravity incorporating a cluster mass of 10^{14} M⊙ at the Einstein radius r_E = 100 kpc, Hubble expansion H(z=0.5), a lensing efficiency parameter L(t), and the Aether EM correction. The result, g_Rings ≈ 1.053×10^{-2} m/s^2, represents the effective infall acceleration at the Einstein radius.

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1. Introduction

Strong gravitational lensing by massive clusters provides a direct measurement of projected mass within the Einstein radius. For GAL-CLUS-022058s the Einstein radius corresponds to a physical scale of ~100 kpc at the lens redshift z = 0.5. UQFF adds three corrections to the standard lensing formula:

1. Hubble correction at z = 0.5: $H(z) = H_0\sqrt{}(\Omega\_m(1+z)^3 + \Omega\_\Lambda)$
2. Lensing efficiency term: $L(t) = GM/(c^2\cdot r) \times D\_LS/D\_S$
3. EM Aether term: $q \times (v \times B) \times 11 \times 10^{-12}$

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2. Master UQFF Gravity Equation

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g_Rings(r_E, t) = [G·M_cluster / r_E²] × (1 + H(z)) × (1 — B/B_crit)
                × (1 + L(t))
                + q·(v_ICM × B_ICM) × A_aeth × A_scale

H(z=0.5) = H_0 × sqrt(Ω_m × (1+z)³ + Ω_Λ)
         = 70 × sqrt(0.3 × 3.375 + 0.7)
         = 70 × sqrt(1.7125) ≈ 70 × 1.3086 ≈ 91.63 km/s/Mpc

L(t) = [G·M / (c²·r_E)] × (D_LS / D_S)
     = 6.674×10⁻¹¹ × 1.989×10⁴⁴ / (9×10¹⁶ × (3.086×10²⁰)²) × 0.5
     ≈ 2.388×10⁻⁴
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3. Parameters

Parameter	Symbol	Value	Unit
Cluster mass	M	1.989×10^{44}	kg (10^{14} M⊙)

Parameter	Symbol	Value	Unit
Einstein radius	r_E	3.086×10^{20}	m (100 kpc)
Lens redshift	z_lens	0.5	—
D_LS/D_S ratio	D_LS/D_S	0.5	—
H(z=0.5)	H_z	91.63	km/s/Mpc
Lensing efficiency	L(t)	2.388×10^{-4}	—
ICM B-field	B_ICM	1.00×10^{-5}	T
ICM velocity	v_ICM	1.00×10^{6}	m/s
Aether factor	A_aeth	11	—
Scale factor	A_scale	10^{-12}	—
Evaluation epoch	t	5 Gyr	—

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4. Numerical Result (t = 5 Gyr)

$$g\_grav = G \times 1.989\times10^{44} / (3.086\times10^{20})^2$$
$$= 6.674\times10^{-11} \times 1.989\times10^{44} / 9.523\times10^{40}$$
$$\approx 1.394\times10^{-7}\ \mathrm{m/s^2} \quad \text{[bare gravity – small]}$$

$$\times (1 + H\_z) \times (1 + L) \approx \times 1.0003 \approx \text{same}$$

$$g\_EM\ (\text{dominant}) \approx 1.053\times10^{-2}\ \mathrm{m/s^2}$$

$$g\_Rings \approx 1.053\times10^{-2}\ \mathrm{m/s^2}$$

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5. Einstein Ring Geometry

$$\theta\_E = \mathrm{sqrt}(4GM / (c^2 \times D\_eff)) \quad \text{[Einstein angle]}$$
$$D\_eff = D\_L \times D\_S / D\_LS$$

$$\text{Magnification: } \mu = \theta\_E / \Delta\theta \quad \text{[for point source]}$$
$$\text{Source arc length: } l = 2\pi \times D\_L \times \theta\_E$$

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6. Conclusions

The Molten Ring (GAL-CLUS-022058s) UQFF model yields $g \approx 1.053\times10^{-2}$ m/s^2 at the Einstein radius, dominated by the Aether EM correction at ICM plasma velocities. The lensing efficiency parameter $L = 2.388\times10^{-4}$ and Hubble correction H(z=0.5) = 91.63 km/s/Mpc are consistent with HST photometric model fits. PAPER_758, CP4 class #342. v5.39.